

Laboratory 1: Basic Arduino and M5stack

Objectives

1. Setup environment, drivers, Arduino IDE, and board libraries.
2. Able to use Arduino IDE with the M5Stack development kit.
3. Able to write program to show characters on the M5Stack LCD and Serial Monitor.

Arduino Platform

There are several platforms available for Internet of Things (IoT) devices. We carefully selected the Arduino platform, because it is ease and inexpensiveness. In particular, “**Arduino is open-source electronics prototyping platform based on flexible, easy-to-use hardware and software. It's intended for making interactive projects.**”¹

Arduino IDE Software

Arduino Software is available for free from <http://arduino.cc/en/Main/Software>. Given that the Arduino tool is created in Java, the software can run on any platform that supports Java. These platforms include Windows, macOS and Linux. (Illustration 1 shows a screenshot of Arduino software running on macOS).

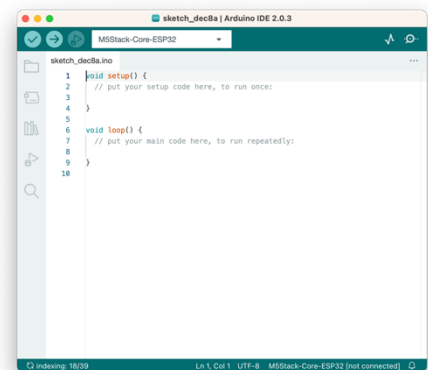


Illustration 1: Screenshot of Arduino Tool

Sketch

A program is called a sketch² in Arduino (with extension **.ino**). It is the unit of code that is written and uploaded to the Arduino board. Basically, every program written for Arduino is a sketch. It is worth clarifying that only one sketch can be uploaded and run on an Arduino board at a particular time. Uploading a new sketch will replace the existing sketch running on the board. The following is the content of a sketch. It must have **setup()** and **loop()** functions.

```
// the setup routine runs once
void setup() {
}

// the loop routine runs over and over again forever:
void loop() {
}
```

¹The definition is taken from <http://www.arduino.cc>

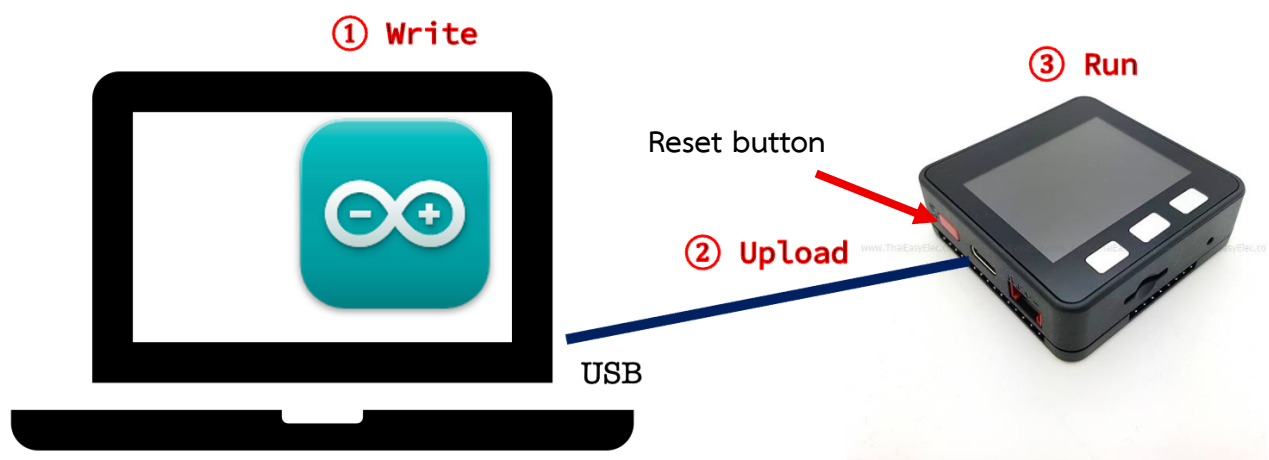
²See <http://arduino.cc/en/Tutorial/Sketch> for more details.

M5Stack

The Arduino development kit (board) that we will use is M5Stack. M5Stack is a robust, open-source development kit with stackable modules, enabling rapid and high-quality prototyping. The core module contains ESP32, 2.0-inch TFT-LCD 320x240 display, 3 buttons, 1 W speaker, I/O extension, TF-card, i2C bus etc.



Working Environment



Workflow

1. Write a sketch in Arduino IDE.
2. Upload to M5Stack.
3. Once upload is completed, M5Stack automatically runs the program. To re-run the program press **RESET** button (red button near USB port)

Lab Exercises

Save each task into separated sketch, i.e., **lab_1_1.ino**, **lab_1_2.ino**.

Task 1: Set up your Arduino development platform for your M5Stack

1.1 Setup Arduino IDE

Visit <https://www.arduino.cc/en/software>. Download and install the latest version of Arduino IDE for your platform (Windows/MAC/Linux). **VERY IMPORTANT!!! You need to install into local drive. DO NOT install to cloud drive, i.e., onedrive, google drive.**

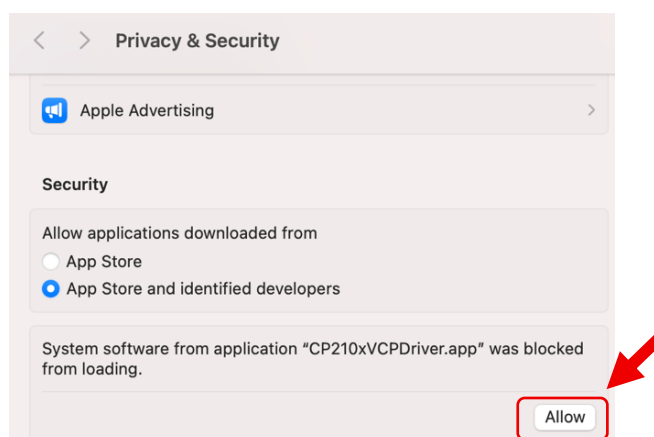
1.2 Setup USB driver

Visit <https://www.silabs.com/software-and-tools/usb-to-uart-bridge-vcp-drivers>, and download the driver for your platform as shown below.

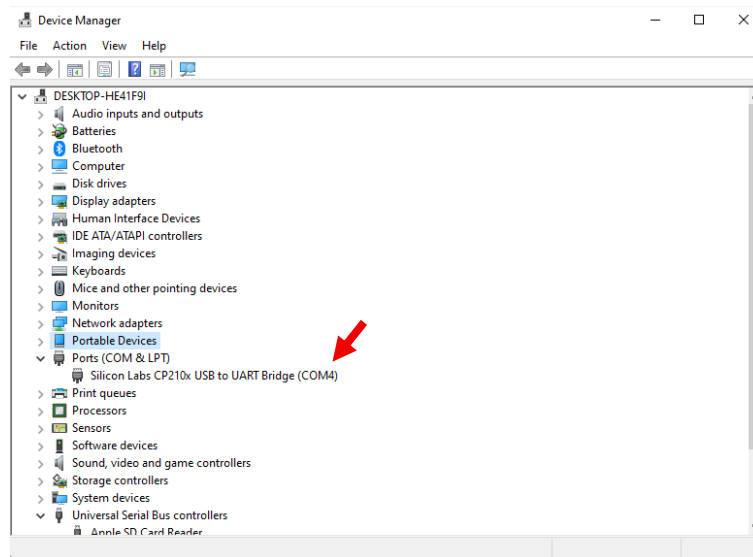
Save **CP210x VCP driver** (zip file) for your platform to a folder. **Do not double click the file.** Extract the zip file and run the installer program.

Software · 11		
Windows	CP210x Universal Windows Driver	v11.5.0 12/31/2025
	CP210x VCP Mac OSX Driver	v6.0.3 5/30/2025
MacOS	CP210x VCP Windows	v6.7 9/3/2020
	CP210x Windows Drivers	v6.7.6 9/3/2020
	CP210x Windows Drivers with Serial Enumerator	v6.7.6 9/3/2020

For Mac user, MacOS might block the installation. You need to open **System Settings > Privacy & Security** and click **Allow** to continue installation.



For Windows user, open Device Manager to see the COM port assigned with the CP210x driver.

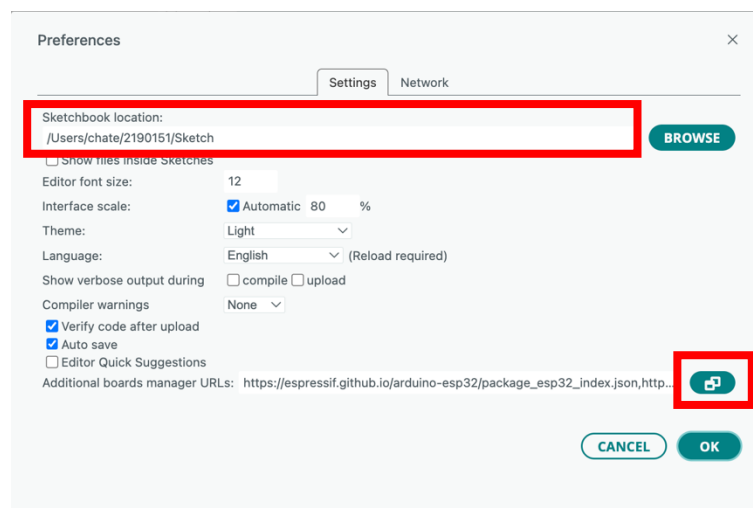


1.3 Setting Arduino IDE Preferences

Open the Arduino IDE. Go to the menu

Mac: *Arduino IDE > Settings...*

Windows: *File > Preferences...*



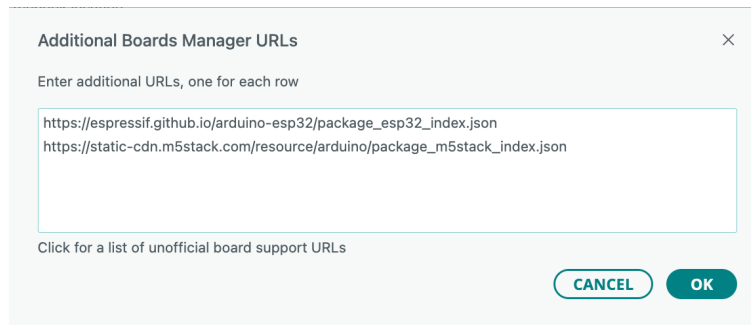
Open file explorer and **create a folder in your local hard drive**. (You need to remember this folder location).

In Sketchbook location, **choose the folder you just created**. This folder will be used for storing all your sketches and installed libraries.

Scroll down to **Additional Board Manager URLs**; click on  and insert the URLs

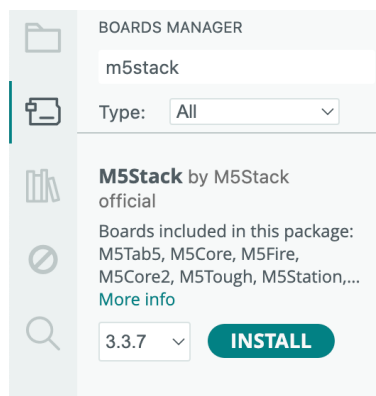
https://espressif.github.io/arduino-esp32/package_esp32_index.json
https://static-cdn.m5stack.com/resource/arduino/package_m5stack_index.json

then click **OK**.

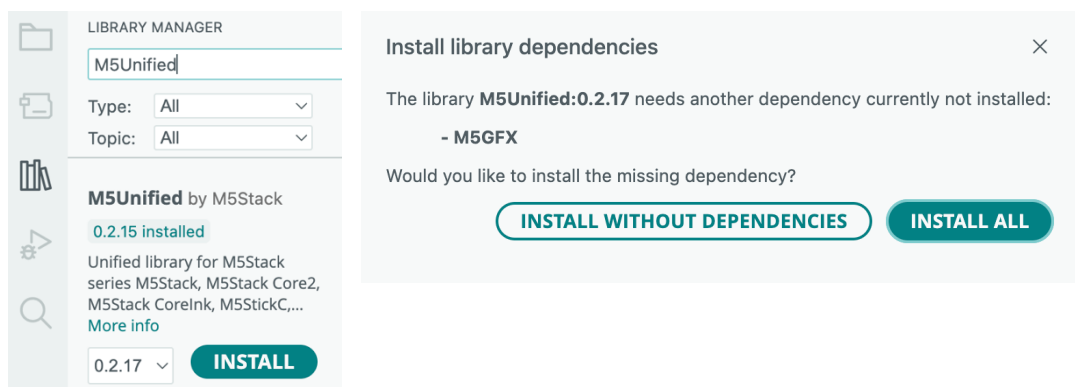


1.4 Setup board and library

- 1.4.1 Go to the menu **Tools > Boards > Boards Manager...**, put “m5stack” in the search box and install the latest version of “**M5Stack** by M5Stack official”.



- 1.4.2 Go to the menu **Sketch > Include Library > Manage Libraries...**, put “M5Unified” in the search box and install “**M5Unified** by M5Stack”. If you see “Install library dependencies” dialog, choose **INSTALL ALL**.



1.5 Setup USB connection

Connect your M5Stack to your laptop computer using the USB cable. In the Arduino IDE, select M5Stack as your board by going to the menu **Tools > Boards > M5Core** and choose the appropriate port in **Tools > PORT** (COM for Windows and UART for Mac) connecting your

computer to M5Stack. For Windows users, you can see the right COM port by inspecting Ports in the **Device Manager**. If everything is correct, you should see message:

M5Core on /dev/cu.SLAB_USBtoUART ([Mac](#)), or

M5Core on COM3 ([Windows](#), if connected to COM3 port),

in the status bar at the bottom of Arduino IDE. This will confirm that Arduino IDE can communicate with M5Stack via USB cable. *If this step is not successful, first check the USB connection, reconnect. If it is not successful, you may need to uninstall and reinstall different versions of the driver (especially for Windows users) until you get to the one that will work for your system.*



1.6 Test with example: HelloWorld

1.6.1 Create your first Arduino sketch by going to the menu **Files > New** and type the following sketch:

```
#include <M5Unified.h>  
  
/* the setup routine runs once when  
   M5Stack is started or reset */  
void setup() {  
  M5.begin(); // initialize M5Stack  
  Serial.begin(115200); // initialize serial communication  
  M5.Display.setTextSize(3);  
  M5.Display.setCursor(50, 50);  
  M5.Display.setTextColor(TFT_YELLOW);  
  M5.Display.print("Hello"); // show text on LCD display  
  delay(2000); // wait 2 seconds  
  M5.Display.print("2190151");  
  delay(1000); // wait 1 second  
  M5.Display.setTextColor(TFT_GREEN); // change text color  
  M5.Display.println("Lab 1");  
  M5.Display.println("Task 1");  
  Serial.println("Lab 1 - Task 1"); // show text on serial monitor  
}
```

```
// the loop routine runs over and over again forever
void loop() {
}
```

1.6.2 Select menu **Files > Save as...** and enter **lab_1_1**, then click **Save**.

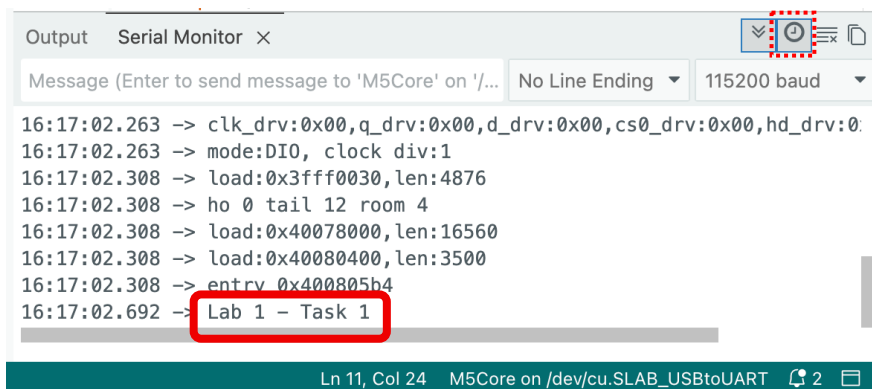
1.6.3 Click the **Upload** button  to flash the M5Stack. This will compile the sketch and upload to M5Stack board if no compilation error. You should see the result as show below. Observe how `print()` and `println()` show output to LCD.



1.6.4 Click Serial Monitor icon.



Then, rerun the sketch by pressing RESET button on M5Stack (red button). Observe the output on LCD and serial monitor. You should see “Lab 1 – Task 1” shown in the Serial Monitor. Click the clock icon to toggle timestamp display.



1.7 Your M5Stack setup is completed.

Task 2: Display text to LCD (lab_1_2)

TODO: Modify the code from Task 1 to make the LCD screen of M5Stack shows your names and IDs of all your group members on separate lines with different colors on the LCD screen with text size setting to **3**. The serial monitor should show the names and IDs in separated line, too. Then, save as **lab_1_2**.

Command references:

- Show text:

`M5.Display.print()` – show text

Color	Hex Value	Red (R)	Green (G)	Blue (B)
TFT_BLACK	0x0000	0	0	0
TFT_NAVY	0x000F	0	0	128
TFT_DARKGREEN	0x03E0	0	128	0
TFT_DARKCYAN	0x03EF	0	128	128
TFT_MAROON	0x7800	128	0	0
TFT_PURPLE	0x780F	128	0	128
TFT_OLIVE	0x7BE0	128	128	0
TFT_LIGHTGREY	0xD69A	211	211	211
TFT_LIGHTGRAY	0xD69A	211	211	211
TFT_DARKGREY	0x7BEF	128	128	128
TFT_DARKGRAY	0x7BEF	128	128	128
TFT_BLUE	0x001F	0	0	255
TFT_GREEN	0x07E0	0	255	0
TFT_CYAN	0x07FF	0	255	255
TFT_RED	0xF800	255	0	0
TFT_MAGENTA	0xF81F	255	0	255
TFT_YELLOW	0xFFE0	255	255	0
TFT_WHITE	0xFFFF	255	255	255
TFT_ORANGE	0xFDA0	255	180	0
TFT_GREENYELLOW	0xB7E0	180	255	0
TFT_PINK	0xFE19	255	192	203
TFT_BROWN	0x9A60	150	75	0
TFT_GOLD	0xFEA0	255	215	0
TFT_SILVER	0xC618	192	192	192
TFT_SKYBLUE	0x867D	135	206	235
TFT_VIOLET	0x915C	180	46	226
TFT_TRANSPARENT	0x0120			

`M5.Display.println()` – show text and move to next line

- Set text size and color:

`M5.Display.setTextSize(textSize);`

`M5.Display.setTextColor(textColor);`

`M5.Display.setTextColor(textColor, backgroundColor);`

`textSize` is an integer number range from 1 (smallest) to 10 (largest).

`color`, `textColor` and `backgroundColor` are shown in Table 1.

Table 1 - Color name for textColor and backgroundColor

Task 3: Dynamic Display (lab_1_3)

3.1 Create new sketch and copy the following code.

```
#include <M5Unified.h>

int n;
void setup() {
  M5.begin(); // initialize M5Stack
  Serial.begin(115200); // initialized serial communication
  M5.Display.setTextSize(5);
  M5.Display.setTextColor(TFT_BLUE);
  n = 1;
  Serial.println("Lab 1 - Task 3");
}

void loop() {
  Serial.print(n);
  M5.Display.print(n);
  delay(500);
  n++;
}
```

Tip: Go to menu **Tools > Auto Format** to make the code more readability.

3.2 Upload and observe the output on LCD screen and serial monitor. You should see the number showing in increasing order from **1, 2, 3, ...** in both LCD and serial monitor.

3.3 **TODO:** Modify the sketch to show the number in **separate line in Serial Monitor** and show only **the latest number near the center of LCD**.

Hint:

Observe the LCD output carefully.

The LCD does not automatically erase previously displayed text.

What happens to the previous number when a new number is displayed?

How can you make the LCD show only the latest value?

Command references:

- Clear screen:
`M5.Display.clear();`
`M5.Display.clear(color);`
- Move text position:
`M5.Display.setCursor(xPosition, yPosition);`
xPosition, yPosition are coordinate in LCD, where 0, 0 is the top-left corner and 319, 239 is the bottom-right corner.
- `delay(t)` – make M5Stack wait for t milliseconds

Task 4: Reflection about this lab. See assignment in myCourseVille. (Individual)

**Lab 1: Basic Arduino and M5Stack**

Section: _____ Date: _____

Members

Name: _____ Student ID: _____

Name: _____ Student ID: _____

Name: _____ Student ID: _____

Task		Graded by
1	Setting up your Arduino development platform M5Stack shows “Hello 2190151” on the LCD screen and serial monitor of M5Stack.	
2	Show names and IDs of all group members: - LCD on separate lines with different colors, text size setting to 3. - Serial Monitor shows names and IDs.	
3	Show the running number on a separate line in Serial Monitor and show the latest number near the center of the LCD.	

4. Reflection assignment in myCourseVille (individual).

- 4.1 What is the difference between `setup()` and `loop()`?
- 4.2 Why do we need to upload the sketch after modifying it?
- 4.3 What is the difference between LCD display and Serial Monitor?
- 4.4 How did you make the LCD show only the latest number?